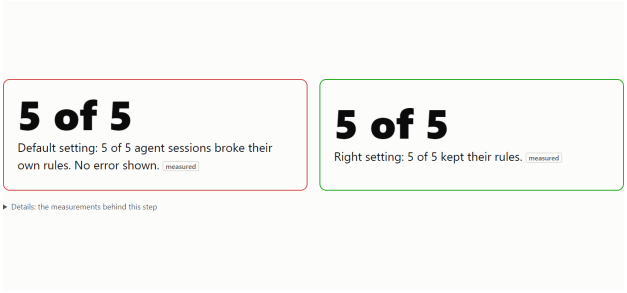
Silent context loss on local AI agents

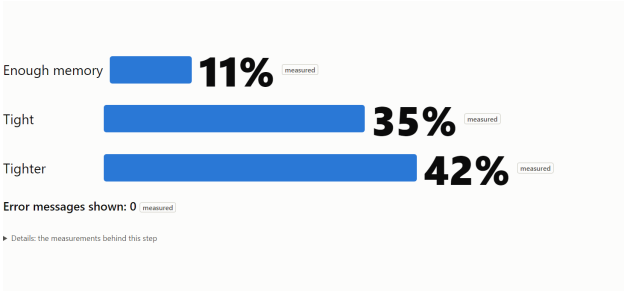
What we measured, what it costs, and the tool that picks the setting

Rithwik Sharma • DSE demo, 28 October 2026 • labels: measured run on our hardware | priced, not executed token counts at list prices, no API call | projected projected from another device or task size



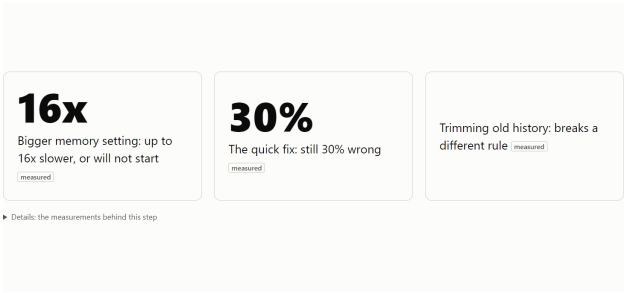
1. Same laptop, two settings

- measured Default setting: 5 of 5 agent sessions broke their own rules. No error shown.[1]
- measured Right setting: 5 of 5 kept their rules.[2]



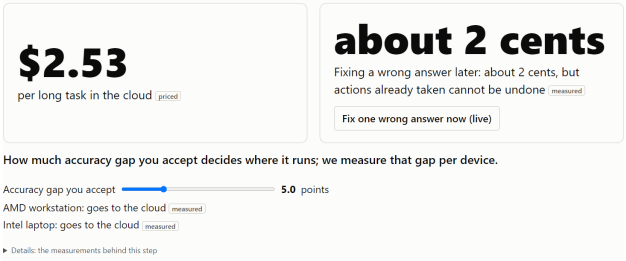
2. How often answers are silently wrong

- measured Enough memory: 11% wrong[3]
- measured Tight: 35% wrong[3]
- measured Tighter: 42% wrong[3]
- measured Error messages shown: 0[3]



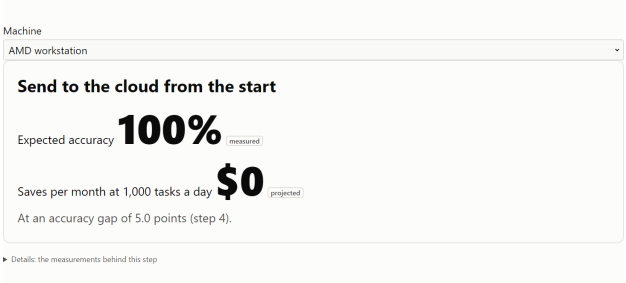
3. Why not just change one setting?

- measured Bigger memory setting: up to 16x slower, or will not start[4][5]
- measured The quick fix: still 30% wrong[3]
- measured Trimming old history: breaks a different rule[6]



4. What it costs

- priced $2.53 per long task in the cloud[7]
- measured Fixing a wrong answer later: about 2 cents, but actions already taken cannot be undone[8]
- How much accuracy gap you accept decides where it runs; we measure that gap per device.



5. What our tool does

- measured AMD workstation: send to the cloud from the start at an accuracy gap of 5 points[9]
- measured Intel laptop: send to the cloud from the start at an accuracy gap of 5 points[9]
- not measured NVIDIA laptop

Still being measured.

- On the Intel laptop, local accuracy is still measured as the share of agent sessions that kept their rules; task accuracy is measured in its next test window (26 October to 8 November).
- The cost of fixing answers with the cheaper cloud model is interim: the fix-later test ran on the top cloud model only.
- The NVIDIA laptop is not measured yet; the tool says so.

Sources (row ids in docs/NUMBERS_REGISTER.md, or result file and line): 1 `t2s-r2-real-cpu`; 2 `t2s-r2-real-igpu`; 3 `x2-evalset-tasks-real-v2-wrong-rate`; 4 `impact-session-time-by-window`; 5 `fh1-amd-boundary-qwen3-32b`; 6 `R2-qwen3-14b-4096-reconciliation`; 7 `cloud-cost-per-real-task`; 8 `cloud-redo-full-context`; 9 `hybrid-route-at-start`.